

DATA ANALYTICS – PROJECT 3

SQL Data Analysis

Intern Details

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Organization: DecodeLabs

Project: Project 3 – SQL Data Analysis

Objective

The objective of this project is to analyze transactional data using SQL queries. The project focuses on extracting meaningful insights through data filtering, grouping, sorting, and aggregation.

Dataset Overview

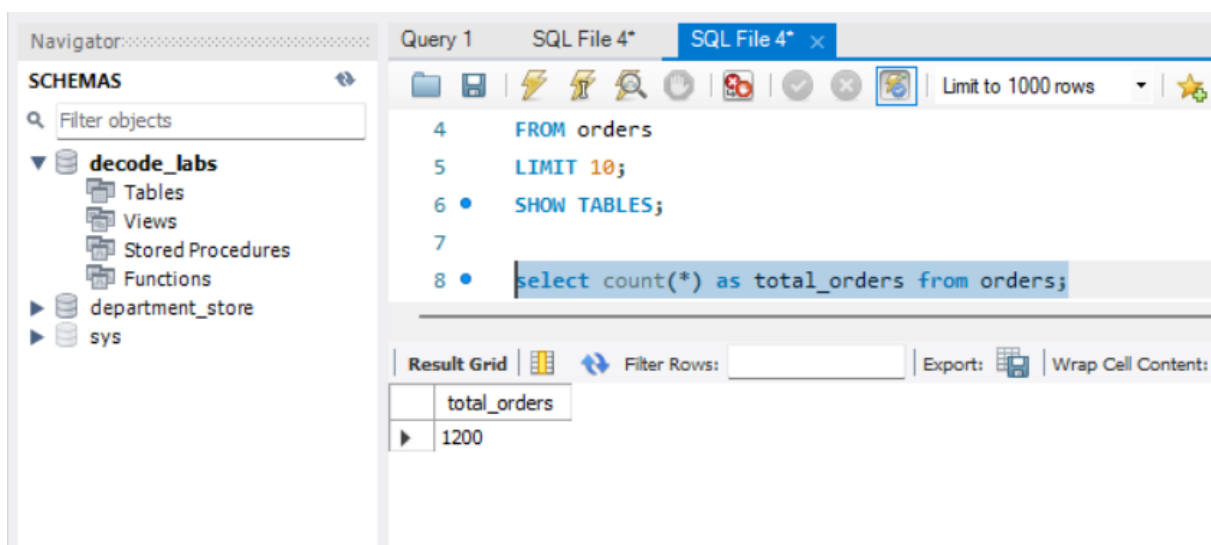
The cleaned dataset from previous projects was imported into MySQL Workbench and analyzed using SQL.

SQL Queries Performed

Query 1: Total Orders

Used COUNT() to determine the total number of records.

Screenshot:



Query 2: Average Order Value

Used AVG() to calculate the average transaction value.

Screenshot:

The screenshot shows the SQL Server Enterprise Manager interface. On the left, the 'SCHEMAS' pane is open, showing the 'decode_labs' database with 'Tables', 'Views', 'Stored Procedures', and 'Functions'. The 'department_store' and 'sys' schemas are also visible. The main pane displays 'Query 1' with the following SQL code:

```
7  
8 • select count(*) as total_orders from orders;  
9  
10 • select Avg(TotalPrice) as Total_Average from orders;
```

The 'Result Grid' is shown below the query, displaying the following data:

Total_Average
1053.9683000000005

Query 3: Maximum Order Value

Used MAX() to identify the highest-value transaction.

Screenshot:

The screenshot shows the SQL Server Enterprise Manager interface. On the left, the 'SCHEMAS' pane is open, showing the 'decode_labs' database with 'Tables', 'Views', 'Stored Procedures', and 'Functions'. The 'department_store' and 'sys' schemas are also visible. The main pane displays 'Query 1' with the following SQL code:

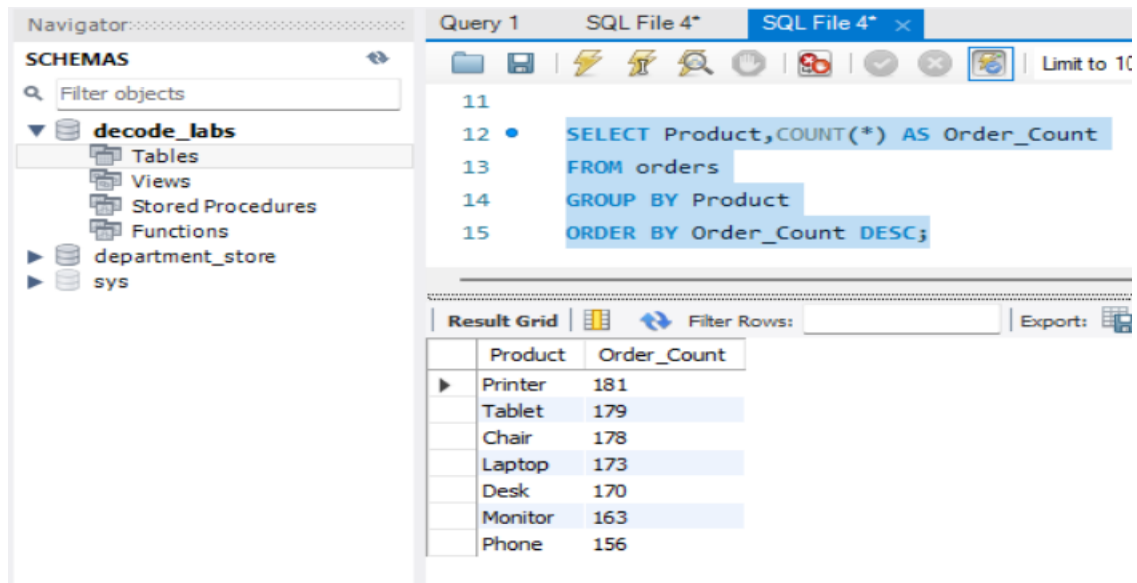
```
9  
10 • select Avg(TotalPrice) as Total_Average from orders;  
11  
12 • select Max(TotalPrice) as Highest_order from orders;
```

The 'Result Grid' is shown below the query, displaying the following data:

Highest_order
3456.4

Query 4: Product Analysis

Used GROUP BY and COUNT() to identify the most frequently ordered products.



The screenshot shows the SQL Server Enterprise Manager interface. On the left, the 'Navigator' pane displays the 'decode_labs' database schema, including tables, views, stored procedures, functions, and the 'department_store' and 'sys' schemas. The main pane shows 'Query 1' with the following SQL code:

```
11
12 • SELECT Product, COUNT(*) AS Order_Count
13 FROM orders
14 GROUP BY Product
15 ORDER BY Order_Count DESC;
```

Below the query editor, the 'Result Grid' tab is active, displaying the results of the query. The results are sorted by 'Order_Count' in descending order.

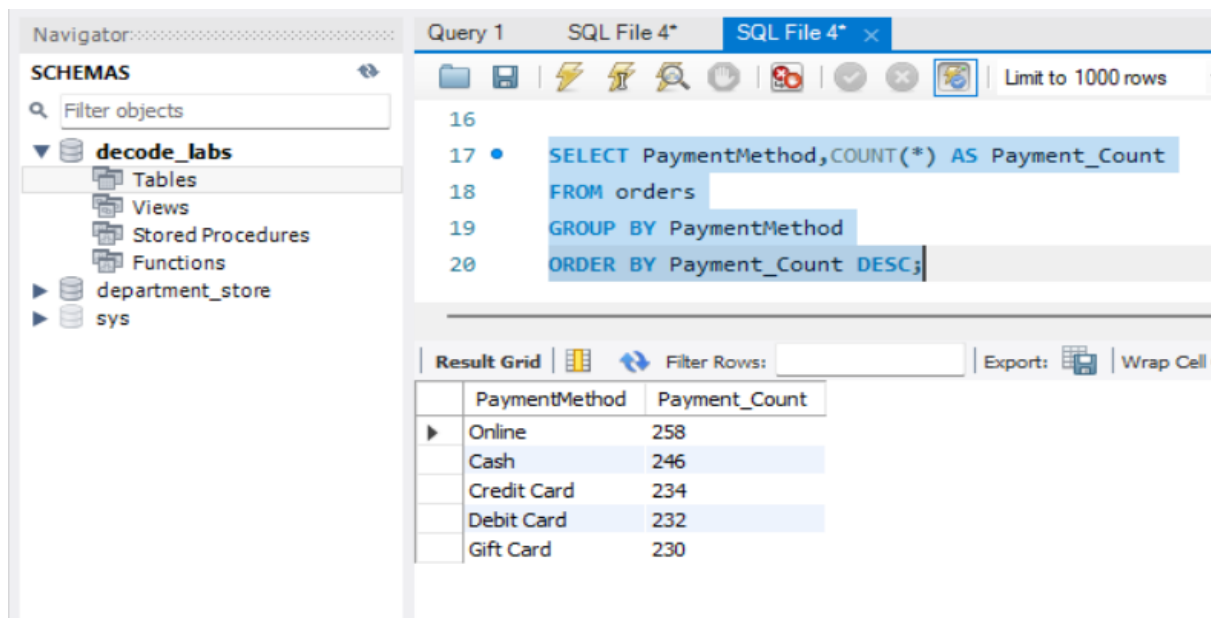
Product	Order_Count
Printer	181
Tablet	179
Chair	178
Laptop	173
Desk	170
Monitor	163
Phone	156

Observation:

Printer was the most frequently ordered product.

Query 5: Payment Method Analysis

Used GROUP BY and COUNT() to analyze payment preferences.



The screenshot shows the SQL Server Enterprise Manager interface. On the left, the 'Navigator' pane displays the 'decode_labs' database schema, including tables, views, stored procedures, functions, and the 'department_store' and 'sys' schemas. The main pane shows 'Query 1' with the following SQL code:

```
16
17 • SELECT PaymentMethod, COUNT(*) AS Payment_Count
18 FROM orders
19 GROUP BY PaymentMethod
20 ORDER BY Payment_Count DESC;
```

Below the query editor, the 'Result Grid' tab is active, displaying the results of the query. The results are sorted by 'Payment_Count' in descending order.

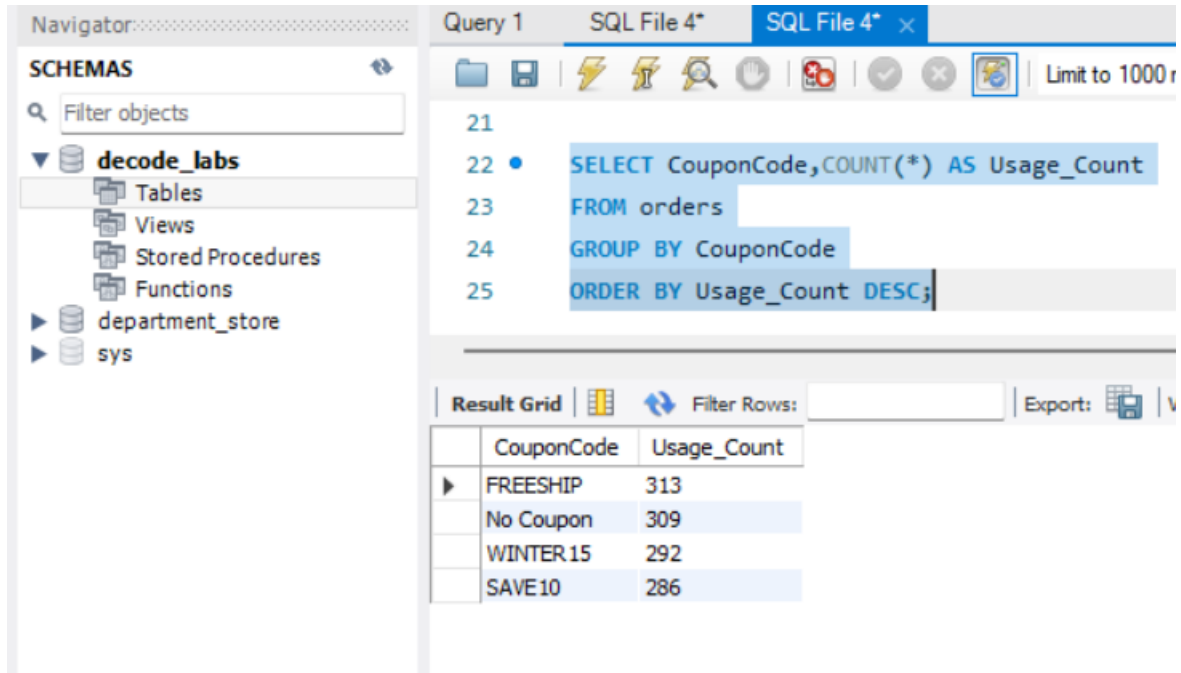
PaymentMethod	Payment_Count
Online	258
Cash	246
Credit Card	234
Debit Card	232
Gift Card	230

Observation:

Online payment was the most preferred payment method.

Query 6: Coupon Usage Analysis

Used GROUP BY and COUNT() to identify coupon usage trends.



The screenshot shows a SQL IDE interface with a Navigator pane on the left and a Query Editor on the right. The Navigator pane displays the 'decode_labs' schema with tables, views, stored procedures, and functions. The Query Editor shows the following SQL query:

```
21  
22 • SELECT CouponCode, COUNT(*) AS Usage_Count  
23 FROM orders  
24 GROUP BY CouponCode  
25 ORDER BY Usage_Count DESC;
```

The Result Grid below the query shows the following data:

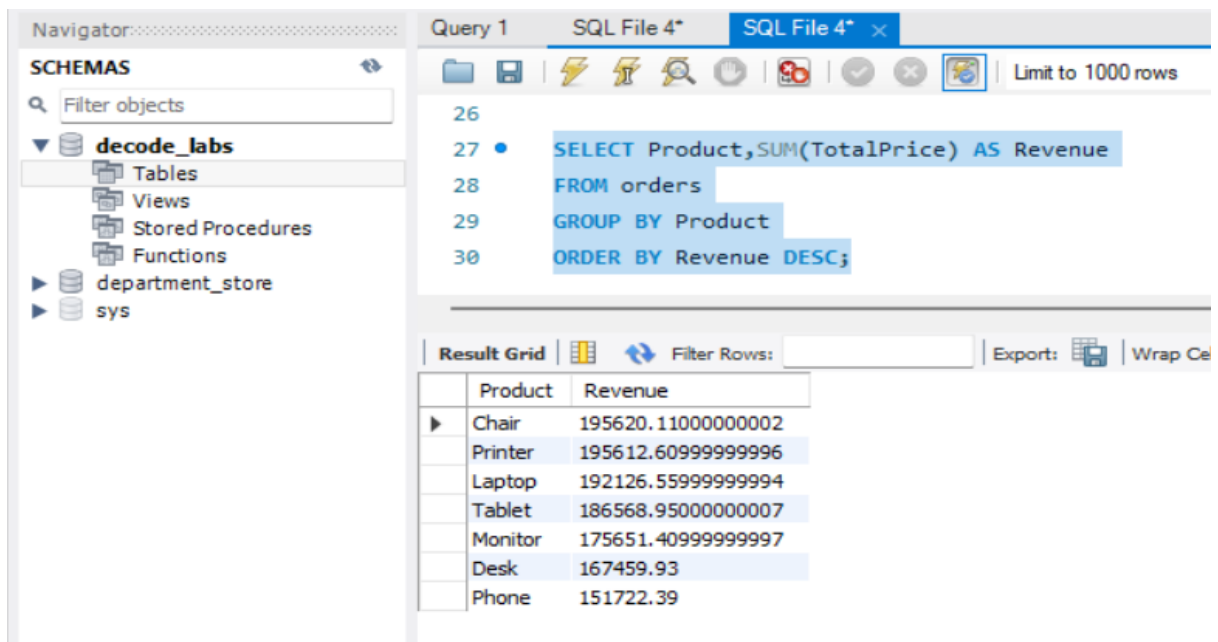
CouponCode	Usage_Count
FREESHIP	313
No Coupon	309
WINTER15	292
SAVE10	286

Observation:

FREESHIP was the most frequently used coupon code.

Query 7: Revenue Analysis

Used GROUP BY and SUM() to calculate revenue by product.



The screenshot shows a SQL IDE interface with a Navigator pane on the left and a Query Editor on the right. The Navigator pane displays the 'decode_labs' schema with tables, views, stored procedures, and functions. The Query Editor shows the following SQL query:

```
26  
27 • SELECT Product, SUM(TotalPrice) AS Revenue  
28 FROM orders  
29 GROUP BY Product  
30 ORDER BY Revenue DESC;
```

The Result Grid below the query shows the following data:

Product	Revenue
Chair	195620.11000000002
Printer	195612.60999999996
Laptop	192126.55999999994
Tablet	186568.95000000007
Monitor	175651.40999999997
Desk	167459.93
Phone	151722.39

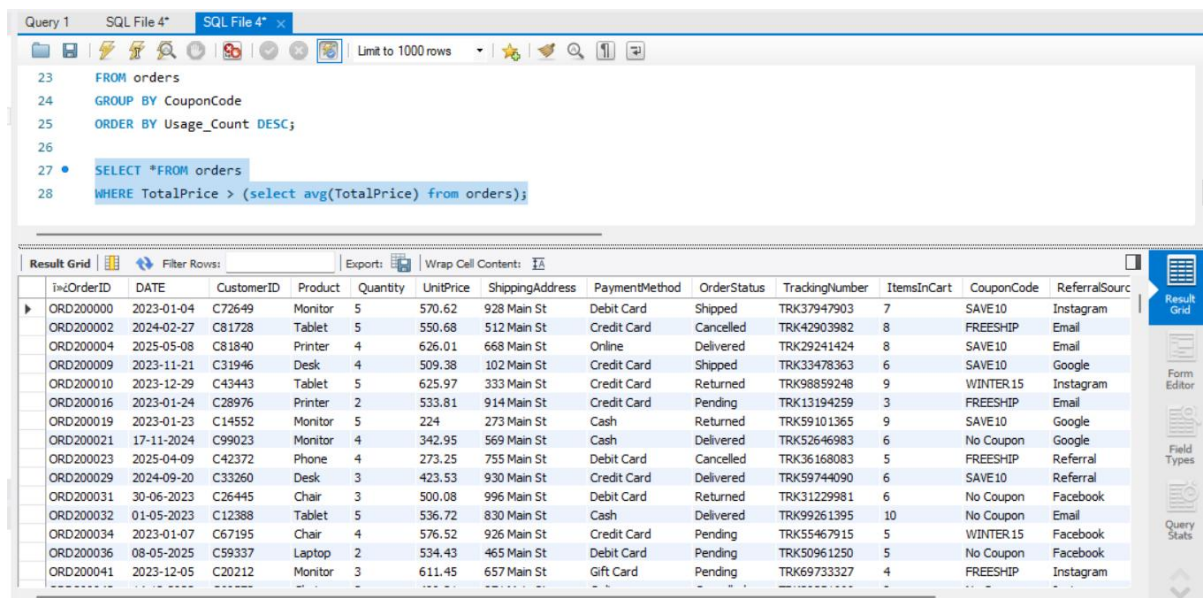
Observation:

Chair generated the highest revenue among all products.

Query 8: Filtered Analysis

Used WHERE clauses to identify:

- Orders above average value
- Online payment transactions
- FREESHIP coupon transactions



The screenshot shows a SQL IDE interface with a query editor and a results grid. The query editor contains the following SQL code:

```
23 FROM orders
24 GROUP BY CouponCode
25 ORDER BY Usage_Count DESC;
26
27 • SELECT *FROM orders
28 WHERE TotalPrice > (select avg(TotalPrice) from orders);
```

The results grid displays a table with the following columns: OrderID, DATE, CustomerID, Product, Quantity, UnitPrice, ShippingAddress, PaymentMethod, OrderStatus, TrackingNumber, ItemsInCart, CouponCode, and ReferralSource. The table contains 16 rows of data, with the first row highlighted.

OrderID	DATE	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	CouponCode	ReferralSource
ORD200000	2023-01-04	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped	TRK37947903	7	SAVE10	Instagram
ORD200002	2024-02-27	C81728	Tablet	5	550.68	512 Main St	Credit Card	Cancelled	TRK42903982	8	FREESHIP	Email
ORD200004	2025-05-08	C81840	Printer	4	626.01	668 Main St	Online	Delivered	TRK29241424	8	SAVE10	Email
ORD200009	2023-11-21	C31946	Desk	4	509.38	102 Main St	Credit Card	Shipped	TRK33478363	6	SAVE10	Google
ORD200010	2023-12-29	C43443	Tablet	5	625.97	333 Main St	Credit Card	Returned	TRK98859248	9	WINTER15	Instagram
ORD200016	2023-01-24	C28976	Printer	2	533.81	914 Main St	Credit Card	Pending	TRK13194259	3	FREESHIP	Email
ORD200019	2023-01-23	C14552	Monitor	5	224	273 Main St	Cash	Returned	TRK59101365	9	SAVE10	Google
ORD200021	17-11-2024	C99023	Monitor	4	342.95	569 Main St	Cash	Delivered	TRK52646983	6	No Coupon	Google
ORD200023	2025-04-09	C42372	Phone	4	273.25	755 Main St	Debit Card	Cancelled	TRK36168083	5	FREESHIP	Referral
ORD200029	2024-09-20	C33260	Desk	3	423.53	930 Main St	Credit Card	Delivered	TRK59744090	6	SAVE10	Referral
ORD200031	30-06-2023	C26445	Chair	3	500.08	996 Main St	Debit Card	Returned	TRK31229981	6	No Coupon	Facebook
ORD200032	01-05-2023	C12388	Tablet	5	536.72	830 Main St	Cash	Delivered	TRK99261395	10	No Coupon	Email
ORD200034	2023-01-07	C67195	Chair	4	576.52	926 Main St	Credit Card	Pending	TRK55467915	5	WINTER15	Facebook
ORD200036	08-05-2025	C59337	Laptop	2	534.43	465 Main St	Debit Card	Pending	TRK50961250	5	No Coupon	Facebook
ORD200041	2023-12-05	C20212	Monitor	3	611.45	657 Main St	Gift Card	Pending	TRK69733327	4	FREESHIP	Instagram

Observation:

SQL filtering enabled targeted analysis of specific customer behaviors and transaction patterns.

SQL Concepts Demonstrated

- SELECT
- WHERE
- ORDER BY
- GROUP BY
- COUNT()
- SUM()
- AVG()
- MAX()

Conclusion

The project successfully demonstrated SQL fundamentals for data analysis. Business insights were extracted using SQL queries, including product performance, payment preferences, coupon usage patterns, and revenue generation. SQL proved to be an efficient tool for transforming raw transactional data into actionable insights.